

Modelling the Impact of COE Quota Changes on COE Premiums

Cat A and B Predictive and Elasticity Analysis (2010 - 2025)

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Overview - Objective

- Predict COE prices for Category A and B using quota and bidding data
- Quantify Elasticity - how COE premiums respond to incremental changes in quota
- Provide a framework for assessing potential policy adjustments

Dataset Overview

- Sources
 - COE Quota (monthly)
 - COE Bidding Results (twice per month)
- Key Variables
 - COE Premium (target variable)
 - Quota (supply)
 - Bids Received / Successful (demand)
 - Derived: demand pressure, lagged premiums & quota, seasonal effects

Feature Engineering

- Engineered Features
 - Demand Pressure = $\text{bids_received} / \text{quota}$
 - Lagged Premium (premium_lag1) to capture market momentum
 - Lagged Quota (quota_lag1) to capture quota change effects
 - Time Features: Monthly seasonality, bidding round (1st/2nd)
 - Category-specific modelling (Cat A and B trained separately)

Predictive Model Approach

- Model Type: Gradient Boosting Regressor
 - Handles non-linear relationships
 - Captures interactions between supply, demand & momentum
 - Robust to noise
- Train/Test Split: 80-20 split, time-series consistent (no shuffling)

Predictive Model Performance

```
===== Training model for Category A =====
MAE: 5,669.15
R² : 0.8450

===== Training model for Category B =====
MAE: 17,156.33
R² : -0.4696
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Category A (More Stable Market)	
Metrics	Value
MAE	\$5,669
R²	0.845
Interpretation	• Good predictability • COE premiums follow strong supply/demand + momentum patterns • Model is suitable for forecasting next bidding round

Category B (Highly Volatile Market)	
Metrics	Value
MAE	\$17,156
R²	-0.47
Interpretation	• Category B prices are extremely volatile and more sensitive to macro shocks, speculative bidding and dealer behavior • Predictive accuracy is lower but still provides directional guidance

Elasticity Model

- Model: Log-log Ordinary Least Squares Regression
 - $\ln(\text{price}) = \beta_0 + \beta_1 \ln(\text{quota}) + \beta_2 \ln(\text{demand pressure}) + \epsilon$
 - Why log-log?
 - Coefficients become elasticities, Direct policy interpretation, Stable event for volatile Cat B

A log-log Ordinary Least Squares (OLS) regression is a model where both the dependent (y) and independent (x) variables are transformed using a logarithmic function. This model is useful for relationships where a **percentage change** in the independent variable results in a **percentage change** in the dependent variable, which can be directly interpreted from the coefficient. The coefficient (β) represents the **elasticity** of y with respect to x .

How it works

- **Model:** The equation is in the form of $\log(y) = \beta_0 + \beta_1 \times \log(x) + \epsilon$.
- **Interpretation:** The coefficient β_1 tells you the **percentage change in y for a 1% increase in x** .
- **Example:** If $\beta_1 = 2$, then a 1% increase in x is associated with a 2% increase in y . If $\beta_1 = -0.5$, a 1% increase in x is associated with a 0.5% decrease in y .

==== Elasticity estimation for Category A ====						
	coef	std err	t	P> t	[0.025	0.975]
const	13.3293	0.227	58.778	0.000	12.883	13.775
log_quota	-0.3659	0.031	-11.694	0.000	-0.427	-0.304
log_dpressure	-0.0735	0.097	-0.760	0.448	-0.264	0.117
==== Elasticity estimation for Category B ====						
	coef	std err	t	P> t	[0.025	0.975]
const	14.4505	0.231	62.642	0.000	13.997	14.904
log_quota	-0.5114	0.034	-15.140	0.000	-0.578	-0.445
log_dpressure	-0.1961	0.104	-1.879	0.061	-0.401	0.009

Elasticity Results (Category A)		
Variable	Coefficient	Interpretation
log_quota	-0.366	1% increase in quota → 0.366% decrease in Cat A COE price
log_dpressure	-0.073	Not statistically significant

Elasticity Results (Category B)		
Variable	Coefficient	Interpretation
log_quota	-0.511	1% increase in quota → 0.511% decrease in Cat B COE price
log_dpressure	-0.196	Demand pressure borderline significant (p≈0.06)

Practical Interpretation of Elasticity

Category A ($\beta = -0.366$)

- +10% quota $\rightarrow$ 3.7% drop in COE price
- If price is \$80K $\rightarrow$ falls by ~\$2,900

Category B ($\beta = -0.511$)

- +10% quota $\rightarrow$ 5.1% drop in COE price
- If price is \$130K $\rightarrow$ falls by ~\$6,600

 Category B responds almost 40% more strongly to quota changes